3 the Method of Maximal Root

and for any $|z_i| \geqslant 1$,

题 3.1. *Let $\{a_n\}$ be a sequence of integers, $z_1, z_2, \cdots, z_k$ are roots of a polynomial $P(x)$ with integer coefficients, $c_1, \cdots, c_k$ are constants such that*

$$a_n - c_1 z_1^n - c_2 z_2^n - \cdots - c_k z_k^n$$

converge to 0 when $n \to \infty$. Prove that for sufficient large n, $\{a_n\}$ satisfies the linear recurrence relation determined by $P(x)$.

题 3.2. (IMOSL2003) *Find all integers $b > 5$, such that the number*

$$x_n = \underbrace{11\cdots1}_{n-1\ 1's}\underbrace{22\cdots2}_{n\ 2's}5$$

under base b is a perfect square for all positive integer n which is sufficiently large.

题 3.3. (IMOSL2005) *Positive integers a, b satisfies that $a^n + n \mid b^n + n$ for any positive integer n, prove that $a = b$.*

题 3.4. (AMM) *Positive integers a, b satisfies that $(a^n - 1) \mid (b^n - 1)$ for all positive integers n, prove that $b = a^k$ for some positive integer k.*

题 3.5. (IMOSL2009) *Let $a \neq b$ be two positive integers greater than 1. Prove that there exists an integer n, such that $(a^n - 1)(b^n - 1)$ is not a perfect square.*

题 3.6. (CTST2025) *Let $P(x), Q(x)$ be non-constant real polynomials, such that for all positive integer m, there exists a positive integer n satisfy $P(m) = Q(n)$. Prove that*

(1) *If $\deg Q \mid \deg P$, then there exists real polynomial $h(x)$ x, satisfy $P(x) = Q(h(x))$ holds for all real number x.*

(2) $\deg Q \mid \deg P$.